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Synopsis

Project Name:

RideMates

Course Code:

CAP463 – CAPSTONE

PROJECT

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1. INTRODUCTION

The modern university ecosystem in Punjab faces a unique logistical challenge: a significant portion of the student and faculty body act as “**day-scholars**,” commuting daily from neighboring cities such as Ludhiana, Jalandhar, and Phagwara. While the distance is manageable (20–40 km), the existing transport infrastructure fails to provide an optimal balance between **cost**, **time**, and **comfort**.

RideMates is a mobile application designed to bridge this gap by creating a **university-exclusive peer-to-peer ride-sharing network**. The platform focuses on “**Asset Utilization**” by connecting vehicle owners (students/staff with cars or two-wheelers) who have empty seats with peers traveling along the same route.

Unlike commercial aggregators like Uber or Rapido, RideMates operates on a **closed-loop verification system** and a **non-profit cost-sharing model**. By restricting access strictly to users with a valid University Institutional Email (e.g., @lpu.in), RideMates creates a secure, legal, and community-driven alternative to public transport without requiring manual ID card checks.

2. MOTIVATION & PROBLEM STATEMENT

The development of RideMates is driven by **three critical inefficiencies** in the current commuting landscape:

2.1 Vulnerability to Strikes & Gridlock

The region frequently experiences disruptions due to:

- Bus Worker Strikes
- Citizen protests (Bandhs)
- Highway gridlocks caused by accidents

During these events, rigid public transport systems collapse, leaving students stranded. **Two-wheelers**, however, offer the necessary agility to navigate alternative routes and link roads, ensuring students can reach the university even when highways are blocked.

2.2 The Hidden Cost of Public Transport

While public bus tickets appear cheap (₹20–30 from Jalandhar, ~₹70 from Ludhiana), the “**Total Commute Cost**” is significantly higher. Students must pay for:

- Auto-rickshaws to reach the bus stand
- Shuttles to reach their campus block

RideMates offers a “**Door-to-Block**” service where the fuel-split cost is comparable to the bus ticket, but eliminates the expensive and time-consuming “**Last Mile**” connections.

2.3 The “Trust Deficit” in Open Networks

Existing carpooling solutions (e.g., BlaBlaCar) and bike-taxis (e.g., Rapido) operate on **open networks**. Students, particularly **female commuters**, are often uncomfortable sharing a ride—especially a two-wheeler—with **unverified strangers** on highways.

3. OBJECTIVES OF THE PROJECT

The primary goal of this project is to develop a **robust Android/iOS application** that democratizes campus commuting.

3.1 Primary Objectives

1. To Create an Automated Verified Ecosystem

Implement a strict approach where every user is authenticated via **Domain-Restricted Email Verification**. Only users who can access a valid @lpu.in inbox can register.

2. To Optimize Two-Wheeler Commuting

Leverage the high density of bikes/scooters to provide an **agile commuting option** that is resilient to traffic jams and strikes.

3. To Implement a Non-Commercial Cost Model

Develop a **“Fair-Share Algorithm”** that calculates fuel costs based on Standardized Vehicle Categories with **vehicle-specific maintenance multipliers**.

3.2 Secondary Objectives

1. Strike Resilience

To enable a **“Route Agility”** feature where drivers can tag rides as **“Emergency/Alternative Route”** during public transport strikes.

2. Social Networking

To transform **dead commute time** into **networking opportunities** between juniors and seniors.

3. Mutual Accountability

To implement a **user-driven Report & Accountability system** that flags no-shows and bad conduct, with **pattern-match evaluation** to penalize repeat offenders and award trust streaks to clean users.

4. PROJECT SCOPE

4.1 Functional Scope

- **User Module**
Automated Registration via University Email OTP (One Time Password), Profile Management.
- **Ride Management**
Post Ride (One-Time), Search Ride, Filter by Visual Vehicle Categories (e.g., “Two-Wheeler”, “Car”).
Map-based Route Visualization (static polyline). Emergency Route Flagging for strike resilience.
- **Safety Module**
“Gender-Lock” (Female-to-Female only filter), “Spare Helmet” Checkbox.
- **Booking Module**
Auto-booking engine with SQL transaction locking to prevent double-booking. Concurrency-safe seat reservation.
- **Report & Accountability Module**
User-filed reports for no-shows and bad conduct; pattern-match evaluation; mutual accountability shield.
- **Post-Booking Handoff**
Native WhatsApp and Call links for off-platform driver-passenger coordination after booking confirmation.

4.2 Exclusions (Out of Scope)

- **In-App Payment Processing**
To avoid banking regulations and complexities, the app will facilitate offline settlements (Cash/UPI) based on the calculated split amount.
- **Real-Time GPS Tracking**
The app matches users based on “Source/Destination” text logic and static route polylines. Live turn-by-turn navigation and vehicle tracking are excluded to reduce battery drain and complexity.
- **In-App Chat/Messaging**
Driver-passenger communication occurs off-platform (WhatsApp, Call) after booking.

5. SYSTEM ANALYSIS & METHODOLOGY

5.1 Technology Stack (React Native + Node.js + MySQL)

Component	Technology	Justification
Frontend	React Native	Chosen for its ability to compile to native Android/iOS code using JavaScript.
Backend	Node.js & Express.js	Chosen for non-blocking I/O, essential for handling concurrent booking requests.
Database	MySQL	A Relational Database Management System (RDBMS) is critical for maintaining structured relationships between Users, Rides, and Trust Scores.
Authentication	Backend-generated OTP (6-digit, delivered via SMTP) + JWT session tokens	Configured to allow sign-ups only from the university domain (e.g., *@lpu.in). OTP expires after 10 minutes; JWT tokens persist for 7 days.

6. FEASIBILITY STUDY

6.1 Technical Feasibility

The project utilizes open-source technologies and demonstrates strong technical feasibility:

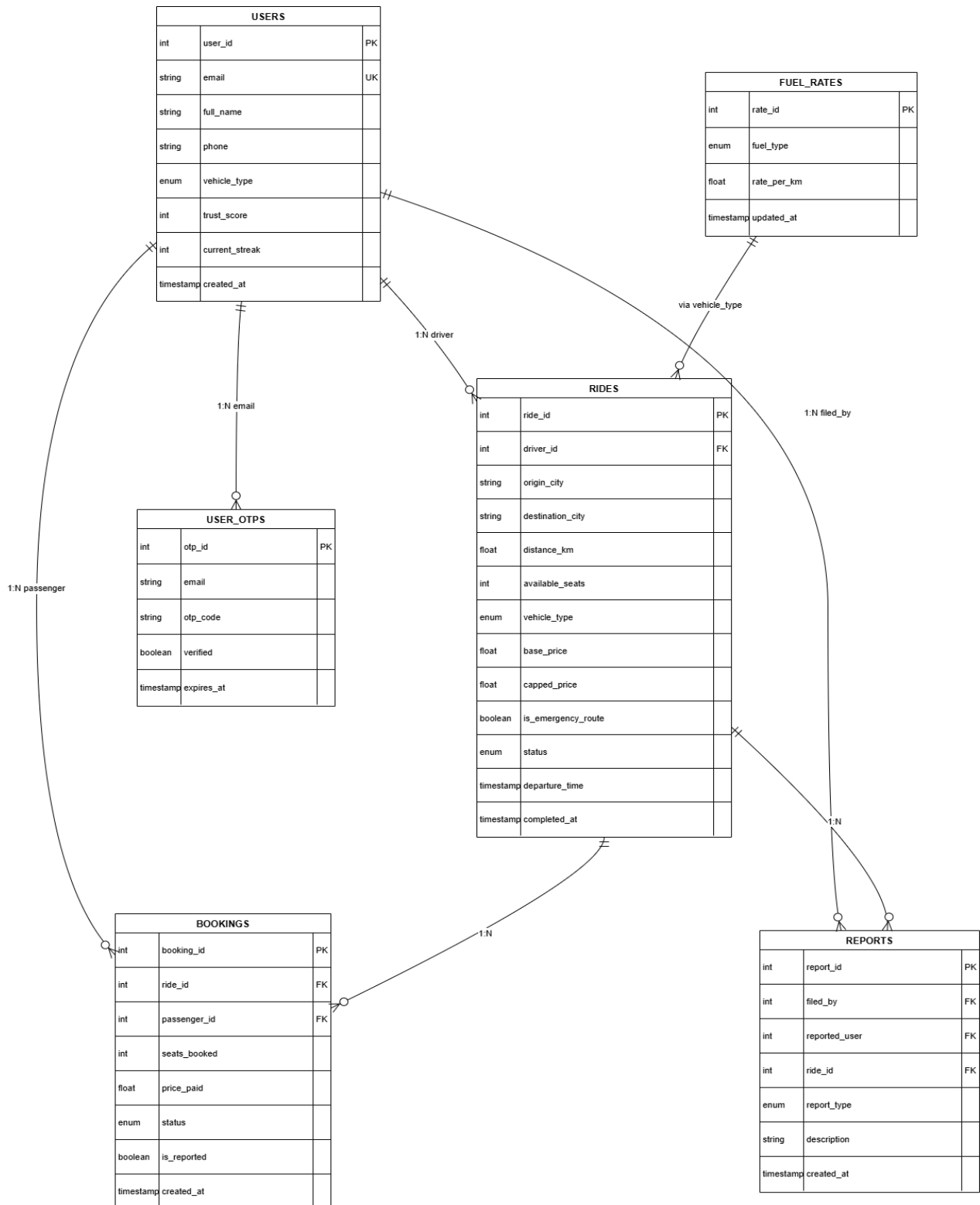
- The team possesses the requisite skills in **React Native**, **Node.js**, **SQL**, and **SMTP integration**.
- Backend OTP generation and verification is **standard practice** in production systems.
- Transaction-level row locking (SELECT...FOR UPDATE) in MySQL ensures **concurrency-safe booking**.
- **Mapbox API** and **SMTP** (e.g., Nodemailer) are battle-tested services for routing and email delivery, reducing development risk.

6.2 Economic Feasibility

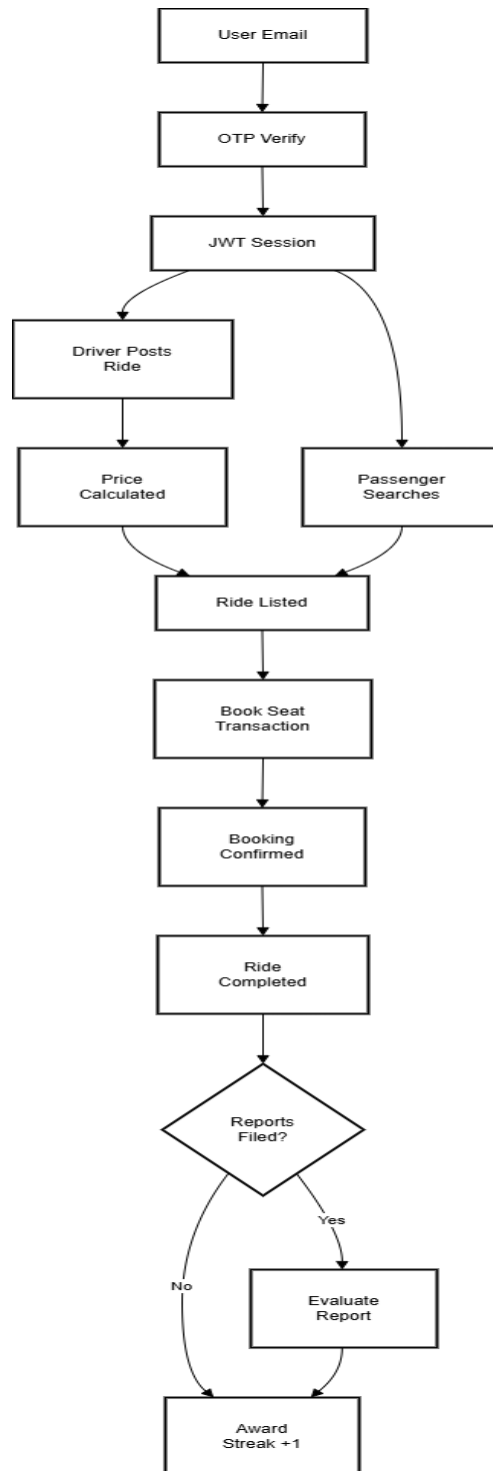
- The project requires zero capital investment.
- Hosting will be managed via **free-tier cloud services**.
- The app relies on the existing hardware (smartphones) of the users.

7. SYSTEM DESIGN (Proposed)

7.1 Database Schema



7.2 Data Flow



8. CONCLUSION

RideMates represents a paradigm shift from “**Transactional Transport**” to “**Community Commuting.**”

By solving the specific pain points of the university demographic—namely:

Cost (inefficient public transport pricing).

Safety (trust deficit in open networks).

Resilience (vulnerability to strikes and gridlock).

RideMates provides a **vital utility** that is:

- **Technically challenging** — Requires expertise in mobile development, concurrent database transactions, and distributed systems
- **Socially responsible** — Fosters peer trust and community engagement
- **Economically viable** — Zero capital cost with sustainable peer-to-peer economics

9. REFERENCES

1. Motor Vehicles Act, 1988 (Section 66)

Regarding private vehicle usage and non-commercial ride-sharing regulations.

2. Case Study: BlaBlaCar

Analyzing the trust model of long-distance carpooling and per-seat pricing mechanisms.

3. React Native Documentation

For cross-platform mobile development standards and best practices.